

IT3060 Human Computer Interaction

Milestones 01-03 (Final Report)

CareQueue

A Patient-Centred Government Hospital OPD Appointment and Queue Management Solution

Group WE_79 / Malabe Campus

REVIEW DRAFT - evidence and member confirmation pending

Prepared 7 October 2026 / Deadline 9 October 2026

Student ID	Name	Proposed M02 workload (unconfirmed actual contribution)
IT23681156	Korala N M	Patient / caregiver / directions
IT23685048	Aluthge D D	Staff queue operations
IT23677296	Withanage A G	Admin / report integration
IT23682764	J.A.I.T. Kalugalla	Accessibility / testing

Executive summary

CareQueue addresses uncertainty around a government-hospital OPD visit: knowing the correct registration point, understanding queue progress, responding to changes and obtaining staff assistance. The design combines patient information with staff control and administrator monitoring. It excludes diagnosis, treatment advice and a complete electronic health record.

The implementation uses React Native, Expo, TypeScript, Firebase Authentication and Firestore. The repository contains 74 current design screens, local demo state, authenticated queue workflows and original design assets. Supplemental management and token-display pages provide record editing, explicit operational decisions, session reporting and assisted access.

This is a review draft. Automated source/domain checks are evidence of implemented logic, not a substitute for installed-device or participant testing. Actual member contributions, five working-app usability sessions, native installation evidence, live account setup and repository/build links must be finalized before submission.

Submission brief: 9 October 2026; maximum 35 main pages including the cover. References and appendices are excluded. The group must demonstrate its own implemented interfaces and explain its decisions.

Milestone 01: problem, scope and stakeholders

The original report describes patients missing announcements, uncertainty about progress, incorrect registration queues, long standing and repeated questions interrupting staff. It proposes shared queue status, estimates labelled approximate, clear directions, authorized clinical priority and continuity during failures.

Primary stakeholders are patients and caregivers, including people with limited digital access. Frontline registration staff, nurses and doctors coordinate registration and clinical flow. Administrators and hospital IT staff monitor disruptions and continuity.

The three personas are Kamala Perera (older patient supported by a daughter), Dilani Fernando (frontline coordinator) and Nuwan Jayasinghe (administrator). They guide priorities; they are design artefacts derived from the supplied records rather than measured app users.

Persona	Key need	Implementation response
Kamala	Readable token; approaching-turn guidance; caregiver help	Queue/status screens; large text; shared visit consent; assisted token display
Dilani	Low-burden registration and safe queue control	Registration review; selected-record edit; call/recovery/priority confirmations
Nuwan	Consistent status and measurable operational context	Shared service state; verified publication; session reports; immutable audit

Milestone 01: research evidence and limits

The Milestone 01 report describes ten supplied structured proxy-response records: five primary, three secondary and two tertiary. Its own limitations say participant authenticity and consent need verification. The following values are reported discovery data, not newly verified observations or population estimates.

Qualitative themes include unclear progress, fragile announcements, physical waiting burden, navigation friction, staff interruption, changing clinical priority and digital exclusion. These themes justify the feature selection without establishing reduced hospital waiting time.

The original appendix includes inconsistent identifiers such as US21 and NFR08-NFR10, while Section 9 defines FR01-FR12 and NFR01-NFR07. This report uses the canonical Section 9 IDs and does not silently expand the requirements.

Reported discovery indicator	Value	Interpretation limit
Primary records waiting at least one hour	4/5	Self-reported small proxy dataset
Approaching-turn clarity mean	2.4/5	Discovery rating, not app usability score
Queue uncertainty	7/10	Record-level theme count
Communication problems	8/10	Record-level theme count
Frequent staff queue questions	5/5 staff/management	Requires original evidence/consent verification

Milestone 02: design development and fidelity

Milestone 02 compares queue-first and task-first patient layouts. The selected home prioritizes active queue information, with preparation and help nearby. Wireframes group token, status, estimate, room and freshness. Review steps precede consequential changes.

The submitted Milestone 02 report contains earlier staff desktop wireframes and loose mobile artwork. The current Figma file uses actual linked frames, mobile staff and mobile administrator screens, with confirmation and result states. Existing runtime assets and screen data were retained rather than replacing screens with flat screenshots.

The current repository inventory is 32 patient, 23 staff and 19 administrator screens. There are 474 prototype connections verified against rendered controls. These counts describe available designs/navigation, not 74 separate CRUD implementations.

The supplied Figma URL points to a broad board. Queue Status node 16:95 was read directly during this review with its reference screenshot. Previous intermediate-stage gaps remain documented; reconstructed sketches must not be described as earlier participant validation.

Technology stack and justification

React Native allows a mobile UI and navigation on Android and iOS from a shared TypeScript codebase. Expo supplies compatible camera, font, notification-permission, print/share and build tooling; it reduces initial native integration work for a student project. Native permission and layout behavior still require physical-device tests.

Firebase Authentication identifies users. Trusted owner-assigned custom claims distinguish patient, staff and administrator privileges. Firestore provides shared documents, realtime listeners and transactions for queue mutations. Security rules protect database access rather than relying only on hidden UI buttons. Firebase documentation supports the authentication/rules model and transaction semantics [R4-R5].

AsyncStorage preserves demo state and device-local manual notes. It is not used as an authoritative shared queue. TypeScript, ESLint and Node tests check types, code quality and operational invariants. EAS preview configuration requests an APK for internal distribution [R3].

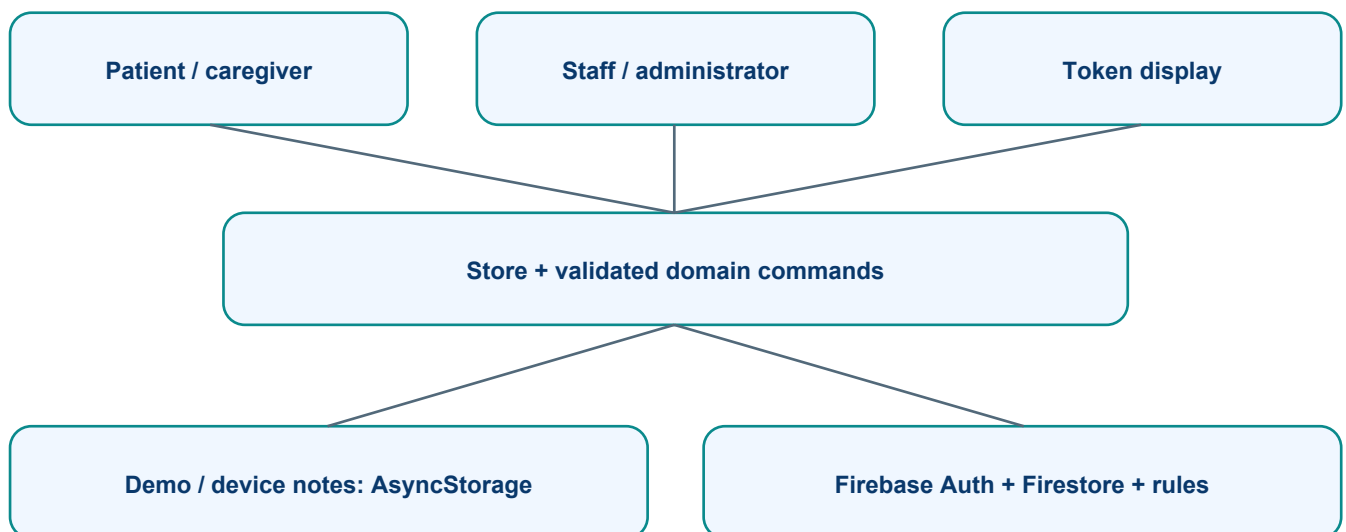
Choice	Requirement fit	Trade-off
React Native / Expo	Runnable mobile app; camera/printing; shared role flows	Native SDK compatibility and device QA
TypeScript	Safer state/command contracts	Types do not validate untrusted remote data automatically
Firebase Auth + claims	NFR07 authenticated roles	Owner must provision privileged accounts
Firestore listeners/transactions	FR01/FR07 shared queue consistency	Transactions fail offline; current single operation document limits scale
AsyncStorage	FR12 continuity and repeatable demos	Notes require human reconciliation; no automatic clinical decisions

Architecture and information boundaries

The patient, staff and administrator screens render the imported design graph. Interaction handlers invoke domain commands through StoreProvider. Demo mode persists a local scenario; Firebase mode subscribes to authenticated service/ticket/operation documents and applies transactions.

Public service state contains current serving token, status and update time. Staff operations contain identifiable registration records; patient profiles contain their own settings and consent. Audit events preserve operational history. Caregiver shares expose only token, visit status and people ahead to an explicitly named account.

Owner role assignment uses the Admin SDK on a trusted machine. Administrator credentials are excluded from the mobile app. The token display renders no patient names, phone numbers or clinical reasons. Current-session report exports also omit these fields.



Implementation: patient and caregiver journeys

Patient screens provide language selection, visit preparation, token retrieval/QR route, queue and timeline information, preferences, approaching/called states, service changes, recovery, directions and help. A normal queue return respects a called or missed visit and pending recovery.

The number ahead now follows actual call order: currently called first, then clinician-prioritized waiting records, then other waiting records. Completed, missed and cancelled visits are excluded. Live estimates use an explicit 3-5 minute heuristic per person ahead; they are not calibrated clinical predictions.

Patients can save, read and revoke caregiver consent. An optional caregiver account ID grants limited shared-visit access. A caregiver signs into a normal account and reads Visits shared with me in the supplemental page. Revocation removes the share. This source path requires new-rule deployment and multi-account verification. Account linking clears an old share when a new visit is linked, so the patient must confirm sharing for that visit.

SMS and background push delivery are not connected. Foreground in-app status is implemented; permission prompts and saved preferences alone are not evidence of external alert delivery. Full Sinhala/Tamil translations remain incomplete.

Implementation: staff operations and continuity

Registration validates a fictional patient name, optional Sri Lankan phone and General OPD service. The first empty live queue now starts at A001; subsequent registrations choose the next number from all retained records. Cancelled visits retain their token/history, avoiding immediate reuse.

Queue operations support selected-record editing, priority with a clinician-reviewed reason, next call/recall, mark-missed, completion and cancellation. Review cards display consequential actions before confirmation. Calls are blocked while the service is delayed or unavailable.

Recovery decisions are explicit approved or rejected outcomes. The transaction re-reads a pending request before deciding, avoiding a stale client-only approval. Account linking prevents assigning an already-linked token to another account or giving one account a second active queue visit.

Manual notes survive locally under the signed-in account ID. Reconciliation requires reviewing notes against current records; it archives them in protected audit before clearing the pending device list. It does not automatically convert an unverified paper note into a clinical action.

Implementation: administration and reporting

Administrators can verify an incident, publish a reviewed service message, change availability and withdraw a message while preserving audit history. The supplemental session report derives registration, waiting, completed, missed, cancelled and measured registration-to-call waiting time from retained queue records.

Measured wait uses actual createdAt and calledAt timestamps and reports the sample count. Missing timing data yields No measured calls instead of an invented average. Delay-update counts are limited to the retained audit snapshot. Historical chart series and other clinic measures are not claimed as live data.

Token slips can be sent to the platform print UI. Reports use native PDF generation/sharing and browser print support. A privacy-preserving display shows current token, service state, freshness and assistance guidance. Real printer, sharing, camera and permission behavior remain device-test items.

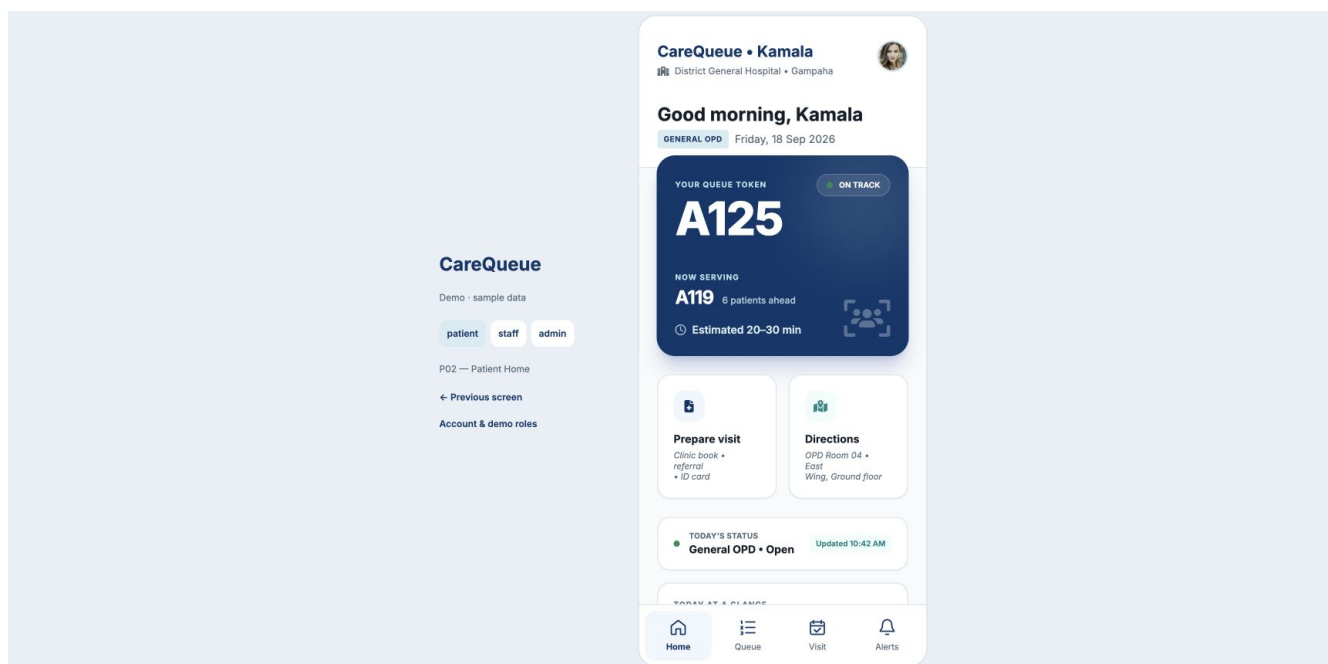
Documented prototype deviations and reasons

The original structure and assets remain the visual reference. The extensions below supply operational behavior that the fixed scenario alone could not demonstrate. Validate the extensions with participants and record any later layout changes.

Change	Reason	Verification / limit
Supplemental My visit / Queue operations pages	Provide live record editing, explicit missed/cancelled/rejected states, sharing and session measures	Domain/rule tests pass; phone interaction testing pending
Header tap opens account as well as long-press	Make account and role access discoverable on a phone	Usability/device evidence pending
Admin is mobile in the current file rather than older report desktop concept	Follow current high-fidelity frame inventory	Design/source mapping; current device fidelity pending
Dynamic queue order and estimated range replace fixed scenario values	Clinical priority and empty queues require truthful computed values	FT01-FT02 pass; heuristic not hospital calibrated
Caregiver account ID links a limited visit share	A phone/name consent alone cannot authorize another app account	Caregiver field/authorization/revocation rules pass
Token-only display and printed slip extension	Support assisted access without exposing clinical/identity details	Source excludes names/phone/reasons; native printing test pending
Immutable audit and retained cancelled visits	Preserve operational history and prevent token reuse	FT01/FT04/FT06 pass; read-only audit is an explicit CRUD coverage limitation
Unconnected delivery/translation/history features remain stated gaps	Avoid treating saved settings or static chart artwork as completed services	Scope resolution and actual implementation/evidence still needed
One raster filename corrected from .png to .jpg	The original file contains JPEG bytes; Android rejects its former extension	Original image bytes and design slots preserved; asset alias prevents re-import regression

Implementation screenshots: archived runtime evidence

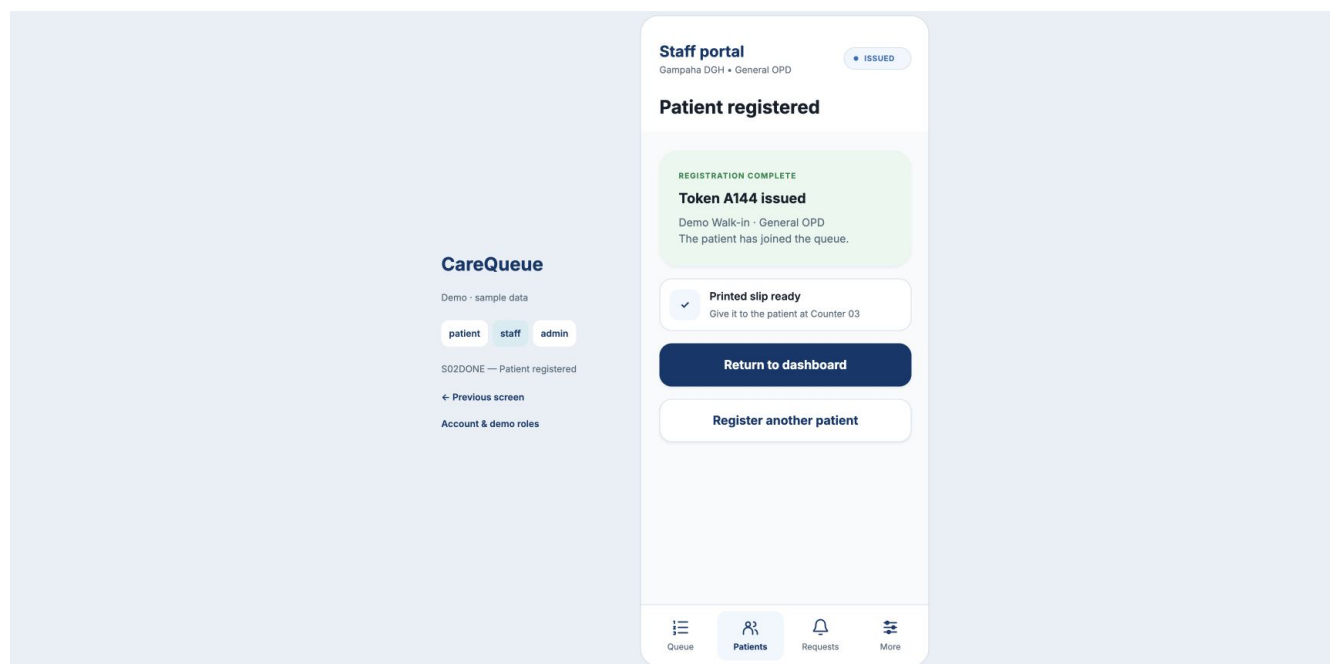
The following screenshot was captured during the repository verification of 5 October 2026. It is an implementation image from the browser demo, not a Figma screenshot and not evidence of the new supplemental features or phone installation. Replace/add current-device evidence after manual testing.



Archived browser implementation evidence / 5 October 2026.

Implementation screenshots: archived staff evidence

This is archived browser-demo staff evidence from 5 October 2026. The runtime register, connected token issuance and audit publication were previously exercised. No physical-device result is inferred from this image.



Archived browser implementation evidence / 5 October 2026.

Traceability: patient requirements

Canonical IDs are from Milestone 01 Section 9. Prototype families refer to Milestone 02/current Figma. Detailed node/source links and all 74 state frames are in traceability.json. Pending evidence is intentionally kept visible.

ID / requirement	Prototype	Implementation / test evidence	Status
FR01 Queue progress and update time	P02/P05/P06; S01/S03; A02	firebase.ts listeners; domain.ts aheadOf; manage.tsx; FT02; W05; MT01	Implemented; connected-device timing pending
FR02 Approximate waiting range	P05/P06/P09/P11	domain.ts waitEstimate; DesignRenderer.tsx; FT02; MT02	Heuristic 3-5 min per person; not hospital calibrated
FR03 Approaching-turn alerts	P07/P09/P10	manage.tsx foreground states; saved preferences; W05; MT03	Foreground in-app states; SMS/background channels pending
FR04 Delay, cancellation, unavailable notices	P11/P12; A04/A05	serviceStatus/cancel commands; shared service listener; FT04; FT06; MT03	App states implemented; external delivery pending
FR05 Registration guidance and directions	P03/P15/P16	DesignRenderer.tsx; display.tsx; token printing; W06; MT04	Scenario directions need hospital confirmation
FR06 Limited authorized caregiver updates	P07/P08; shared visit extension	caregiverShares grant/read/revoke; manage.tsx; W05; caregiver rules case; MT05	Implemented in source; rule rollout and multi-account test pending

Traceability: operational requirements

Canonical IDs are from Milestone 01 Section 9. Prototype families refer to Milestone 02/current Figma. Detailed node/source links and all 74 state frames are in traceability.json. Pending evidence is intentionally kept visible.

ID / requirement	Prototype	Implementation / test evidence	Status
FR07 Walk-in registration and queue control	S01/S02/S03	register/editPatient/callNext/cancel; manage.tsx; FT01; FT03; FT04; W02; MT06	Implemented; concurrent live-device test pending
FR08 Clinician-reviewed priority and reason	S04; A08	priority command; authenticated staff claims; audit; FT02; FT07; W03; MT02	Staff role must represent clinician-reviewed authority
FR09 Missed-turn recovery and decision	P13/P14; S05	requestRecovery; approve/reject transaction; FT05; W01; recovery rules case; MT07	Implemented; real multi-account flow pending
FR10 Flow/wait/delay/missed reports	A01/A02/A03/A06	operationalMetrics; report.ts; manage.tsx; FT08; MT08	Current-session metrics; historical chart pipeline pending
FR11 One verified shared service update	S06; A04/A05; token display	broadcast/withdraw; services/general-opd; display.tsx; FT06; W04; MT08	App/display status share source; external channels pending
FR12 Manual fallback and reconciliation	S07/S08; A07/A08	device notes; explicit reconciliation; audit; FT09; MT09	Notes archived after review; queue corrections remain explicit

Traceability: quality requirements

Canonical IDs are from Milestone 01 Section 9. Prototype families refer to Milestone 02/current Figma. Detailed node/source links and all 74 state frames are in traceability.json. Pending evidence is intentionally kept visible.

ID / requirement	Prototype	Implementation / test evidence	Status
NFR01 Simple task wording	All	Existing labels plus supplemental operations; W08; MT10; T01-T12	Usability evidence pending
NFR02 Accessible/assisted alternatives	P16; display; settings	large text; 52px management controls; token display/slip; MT04; MT10; T12	Screen-reader/device text expansion pending; translations partial
NFR03 Confirmed update visible within 5s	Connected queue views	Firestore realtime listeners; MT01	Not measured on physical devices
NFR04 Freshness and stale indication	Queue/service/health	last-update text; NetInfo disconnected banner; MT09; MT10	Disconnected state implemented; connected stale-age threshold pending
NFR05 Documented manual continuity	S07/S08; A07	manage.tsx guidance and durable notes; FT09; MT09	Physical restart/retry evidence pending
NFR06 Public information boundaries	Token display/slip/report	token-only display; sanitized reporting; caregiver share fields; caregiver rules case; MT04; MT05	Public view renders no record names/phone/reasons
NFR07 Authenticated role restrictions	All protected operations	Custom claims; Firestore rules; route gates; FT07; rules tests; MT10	Owner role assignment; new rule deployment pending

CRUD coverage and individual contributions

The brief requires at least two working CRUD operations per assigned interface. At flow level, registration offers create/read/update, queue control read/update, preferences read/update, caregiver consent create/read/update/delete, recovery create/read/update, broadcasts create/read/delete and manual notes create/read/update.

An outcome screen, navigation transition, exported PDF, immutable audit view or static direction page must not be counted as a second CRUD operation by itself. The full register flags each screen as not individually certified; confirm how the assigned interfaces are grouped before claiming rubric completion.

Milestone 01 provides research responsibilities; Milestone 02 proposes implementation responsibilities. The table below reproduces proposed workload, not verified source authorship. Each member must confirm actual work and demonstrate it independently.

Student ID / name	Milestone 02 proposed workload	Actual implementation contribution
IT23681156 / Korala N M	Patient queue, alerts, caregiver, directions	Await member confirmation
IT23685048 / Aluthge D D	Staff registration, queue and clinical priority	Await member confirmation
IT23677296 / Withanage A G	Administrator interfaces and report integration	Await member confirmation
IT23682764 / J.A.I.T. Kalugalla	Accessibility and testing instruments	Await member confirmation

Functional testing: evidence and remaining execution

Twenty-one domain/source tests passed on 7 October 2026: thirteen new completion cases and eight existing workflow/design cases. They cover empty registration, priority order, selected edit, visit state transitions, recovery decisions, role denial, service restrictions, session metrics, notes/retries, local assets, all 474 connections and visible button labels. Additional checks cover private-update freshness, exact preference payload fields, raster resource signatures and called/ended estimate labels.

The full npm run check completed successfully: TypeScript, ESLint and all 21 domain/source tests. The current Firestore emulator suite also passed all four cases, including caregiver privacy/revocation and repeated recovery only after a staff-marked missed visit. Web, Android and iOS bundles exported successfully. These checks do not establish phone installation, actual external delivery or participant usability.

Functional-tests.md defines MT01-MT10 for connected-device propagation, simultaneous registrations, channel behavior, caregiver authorization/revocation, native exports/scanning, offline restart and accessibility. Their actual-result fields remain NOT EXECUTED. New rules are tested locally but not yet deployed; The final ARM64 APK assembled successfully and its package/signature were verified. Package lk.carequeue.mobile, version 1.0.0, minimum SDK 24, target SDK 36. SHA-256: edb75e8443d002f7786e61708acc90a77c4764530e45b89de93fd506c7a01dfd. The preview uses an Android Debug signing certificate. Installation and physical-device results must still be supplied.

Check	Current evidence	Practical limit
Domain/source suite	21 passed	No phone/network/participant evidence
TypeScript	TypeScript and lint passed	No runtime correctness guarantee
Firestore rules	4/4 passed, isolated demo-carequeue emulator	Includes new caregiver privacy/revocation case
APK / native install	ARM64 APK assembled; package/signature verified; phone install unverified	Export is not an installed application
Five-second propagation	MT01 prepared, not measured	Requires two connected clients and timestamps

Usability: plan, execution and analysis

At least five consenting real or role-informed proxy participants must test the working application. The proposed mix is two patients, one caregiver, one frontline representative and one administrator. A pilot should verify task feasibility on the exact installed version.

Tasks T01-T12 are carried forward from Milestone 02 and adapted to the implementation. The moderator gives goals without naming controls. Record independent/helped/unsuccessful/blocked outcome, seconds, errors/help, ease 1-5, observation and recording timestamp where available. An unavailable implementation path is blocked, not a participant mistake.

No working-app participant results were provided at preparation time. There are five blank session forms and an empty observations JSON. The analysis tool reports zero tested participants and will not manufacture success/time/ease measures.

The M02 body says testing is pending, while Appendix B lists recording links for Patient1, Patient2, Doctor, Nurse and Admin. Resolve this inconsistency by inspecting consent, dates, tasks and the tested artefact. Earlier Figma sessions must not be relabelled as M03 working-app sessions.

Measure	Calculation	Current result
Independent completion	Independent / eligible initial attempts; show blocked separately	Not measured
Time	Median completed attempts, with included sample count	Not measured
Task ease	Median/range of actual 1-5 responses	Not measured
Issue occurrence	Affected / exposed participants per task	Not recorded
Participant minimum	Distinct actual participants with initial attempts	0 provided; minimum 5 required

Issues, changes and retest evidence

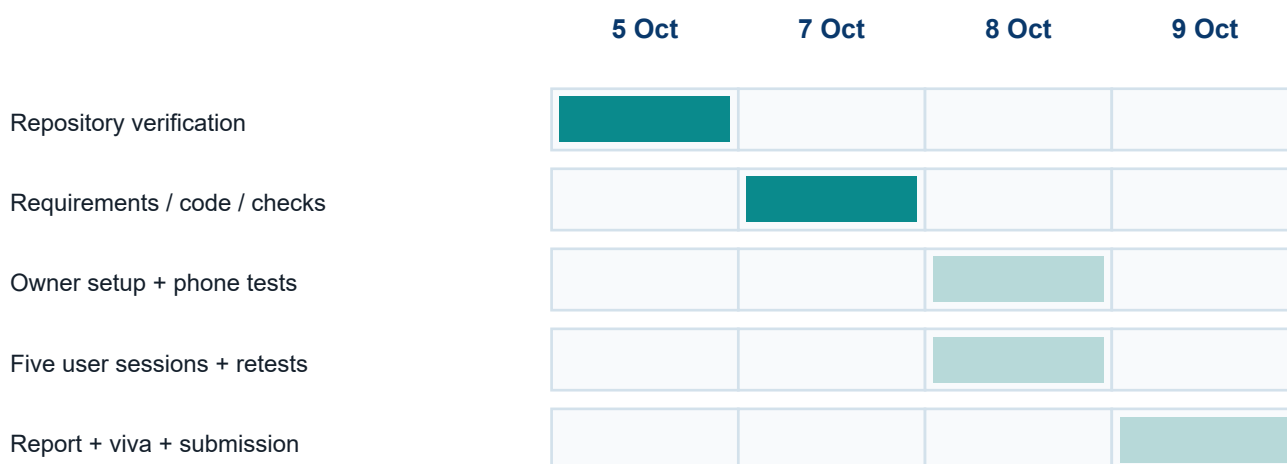
Issue	Action	Evidence / remaining check
Empty live queue can issue invalid token	Use max of valid retained tokens starting at zero	FT01 pass
People ahead ignores clinical priority	Use actual ordered active queue	FT02 pass
Staff cannot edit selected record or distinguish missed/cancelled outcomes	Supplemental management with explicit review	FT03-FT05 pass; device test pending
Old-user asynchronous initialization can affect a new account	Version-guard auth initialization; subscribe after role resolves	TypeScript pass; MT10 pending
Consent does not grant caregiver access	Add limited account-specific share and revocation	New rule test; MT05 pending
Dashboard averages lack measured calls	Derive current-session wait from timestamps/sample count	FT08 pass
Hidden long-press account control	Make header area open account on tap as well	Device usability retest pending
Temporary PDF tools get scanned by project checks	Limit TypeScript source include; ignore tmp/native outputs in lint	Current source typecheck pass
Preference save sends extra app-state fields to protected profile	Project only sms/appAlerts/language/largeText at the write boundary	FT11 and exact-payload emulator test pass
Personal changes misleadingly refresh queue time	Preserve last staff timestamp for private changes	FT10 pass
Called/ended visits display an awaiting-call estimate	Use explicit called/completed/cancelled/missed labels	FT13 pass
JPEG image had PNG filename, blocking Android resources	Rename to .jpg; preserve original bytes and add import alias	FT12 pass; SHA-256 unchanged; stale generated PNG removed

Overall schedule and remaining work

The original milestone reports use relative proposed weeks; they do not establish a complete dated activity log. The chart combines documented repository verification on 5 October with work performed on 7 October and planned completion before the 9 October deadline. Replace earlier milestone dates with actual group records; do not backdate activities.

7 October: reconcile requirements, fix core invariants, prepare traceability and study materials, run source checks, attempt native packaging. 8 October (planned): owner role/rule setup, phone tests, five usability sessions, analyze issues and retest. 9 October (planned): member review, finalized report/links/build, individual viva rehearsal and submission.

The deadline is stated as a date only in the supplied brief. Confirm the submission portal time. Recruitment, consent and individual contribution verification require group action.



Dark teal: recorded activity. Light teal: planned, not completed.

Conclusion and lessons to verify with the group

CareQueue demonstrates the translation of patient uncertainty and staff coordination needs into connected mobile workflows. The implementation preserves clinical decisions with authorized staff, clearly distinguishes approaching/pending/approved states and provides a review path for manual continuity.

Design fidelity alone does not establish operational correctness. Empty data and clinical reordering exposed problems that a fixed prototype scenario would not reveal. Similarly, a saved channel preference does not demonstrate message delivery, and browser screenshots do not demonstrate phone installation.

The group should add its own lessons from implementation and five observed sessions, with concrete examples tied to individual contributions. This draft does not establish measured waiting-time reduction, hospital readiness or complete rubric compliance. Outstanding evidence and scope decisions remain on the submission checklist.

Submission readiness: evidence gates

Before changing this document from review draft to final report, confirm actual workload; verify every assigned interface meets the CRUD requirement; complete at least five working-app usability records; finish device/connected-account checks; verify the supplied APK on a phone and publish reviewed source/repository links; and resolve external-channel/scope limitations.

Add current installed-app screenshots, app build/source hash, functional logs, participant observations/issues and retests to the appendix. Check assessor access to repository and recordings. Keep signed consent, private identities and credentials out of public submission folders.

The brief permits AI tools but expects the group to adapt and own the submitted work. Members must review, verify and rewrite the report in their own voice and explain the implementation. No AI-detection percentage or authorship certification is claimed.

References

[R1] CareQueue group. IT3060HCI2026_Milestone01_GroupWE_79.pdf. Supplied local report, 30 pages; Sections 1-10 and appendices.

[R2] CareQueue group. IT3060HCI2026_Milestone02_GroupWE_79.pdf. Supplied local report, 32 pages; design requirements and planned study.

[R3] Expo. Build APKs for Android devices. <https://docs.expo.dev/build-reference/apk/> (accessed 7 October 2026).

[R4] Firebase. Secure data in Cloud Firestore. <https://firebase.google.com/docs/firestore/security/overview> (accessed 7 October 2026).

[R5] Firebase. Transactions and batched writes. <https://firebase.google.com/docs/firestore/manage-data/transactions> (accessed 7 October 2026).

[R6] CareQueue Figma Design, file OFZrBvawq6lmk96sHEhO88. <https://www.figma.com/design/OFZrBvawq6lmk96sHEhO88/> (current Queue Status inspected 7 October 2026).

[R7] SLIIT IT3060. Assignment 3: Mobile App Implementation & Final Evaluation. Published 28 September 2026; deadline 9 October 2026. Supplied brief, 3 pages.

[R8] CareQueue repository. <https://github.com/xnirupama/CareQueue> . Source, docs/verification.md, archived browser images and current local changes. The new changes must be reviewed and published to the repository before final submission; current remote assessor access is not verified.

Appendix: current screen and CRUD register

This inventory maps all 74 imported state screens to their flow-level operations. It is a coverage audit, not certification of two operations on each state frame. See `traceability.json` and confirmed workload for final assessment.

Node / role	Screen	Flow operations
16:7 / patient	P01 - Language Selection	U: language selection (device state)
16:28 / patient	P02 - Patient Home	R: information/navigation only
16:60 / patient	P03 - Prepare My Visit	R: information/navigation only
16:95 / patient	P05 - Queue Status	R: information/navigation only
16:122 / patient	P07 - Notification Preferences	R/U: preferences saved
16:154 / patient	P08 - Caregiver Authorization	C/R/U/D: caregiver consent; sharing extension
16:189 / patient	P09 - Approaching Turn	R: information/navigation only
16:206 / patient	P11 - Delay Alert	R: information/navigation only
16:227 / patient	P13 - Missed Turn	C/R: recovery request
16:247 / patient	P15 - Directions	R: information/navigation only
16:276 / patient	P16 - Help & Accessibility	R: information/navigation only
18:576 / patient	P04 - Join / Retrieve Queue	R: information/navigation only
18:599 / patient	P06 - Queue Timeline	R: information/navigation only
18:636 / patient	P10 - Your Turn	R: information/navigation only
18:648 / patient	P14 - Recovery Confirmation	R: recovery outcome
135:656 / patient	P12 - Service unavailable	R: information/navigation only
151:769 / patient	P04SCAN - Scan your queue slip	R: information/navigation only
151:835 / patient	P07SAVED - Preferences saved	R/U: preferences saved

Node / role	Screen	Flow operations
153:847 / patient	P08SAVED - Caregiver access	C/R/U/D: caregiver consent; sharing extension
153:919 / patient	P08REVOKED - Access withdrawn	C/R/U/D: caregiver consent; sharing extension
153:987 / patient	P09READY - You're near Room 04	R: information/navigation only
153:1049 / patient	P11SAVED - Updates remain active	R: information/navigation only
153:1117 / patient	P05DELAY - My queue	R: information/navigation only
153:1207 / patient	P15R - Find Counter 03	R: information/navigation only
153:1292 / patient	P16SUPPORT - Talk to OPD staff	R: information/navigation only
158:1140 / patient	P01SI - Sinhala language selected	U: language selection (device state)
158:1207 / patient	P01TA - Tamil language selected	U: language selection (device state)
159:1248 / patient	P07OFF - Notification choices	R/U: preferences saved
159:1314 / patient	P07APPOFF - Notification choices	R/U: preferences saved
159:1380 / patient	P07BOTHOFF - Notification choices	R/U: preferences saved
159:1446 / patient	P07REVOKED - Notification choices	R/U: preferences saved
160:1464 / patient	P05PENDING - Recovery status	R: information/navigation only
138:4 / staff	S01 - Staff dashboard	R/U: queue call
139:223 / staff	S02 - Register a walk-in	C/R/U: registration and review; edit in /manage
139:297 / staff	S03 - Manage the queue	R/U: call/recall; cancel in /manage
139:376 / staff	S04 - Clinical priority	R/U: clinical priority
139:445 / staff	S05 - Missed-turn requests	R/U: recovery decision
140:243 / staff	S06 - Service broadcast	C/R: broadcast
140:1066 / staff	S07 - Offline mode	C/R: manual note extension
140:1131 / staff	S08 - Reconcile offline actions	R/U: reconciliation
140:1208 / staff	SMORE - Staff tools	R: information/navigation only

Node / role	Screen	Flow operations
154:453 / staff	S02REVIEW - Review registration	C/R/U: registration and review; edit in /manage
154:528 / staff	S02DONE - Patient registered	C/R/U: registration and review; edit in /manage
154:605 / staff	S03REVIEW - Call next patient?	R/U: call/recall; cancel in /manage
154:675 / staff	S03CALLED - Patient called	R/U: call/recall; cancel in /manage
154:745 / staff	S03RECALL - Token recalled	R/U: call/recall; cancel in /manage
154:813 / staff	S04REVIEW - Confirm priority change	R/U: clinical priority
154:889 / staff	S04DONE - Priority applied	R/U: clinical priority
154:957 / staff	S05REVIEW - Confirm recovery	R/U: recovery decision
154:1025 / staff	S05DONE - Recovery recorded	R/U: recovery decision
154:1087 / staff	S06REVIEW - Confirm broadcast	C/R: broadcast
154:1166 / staff	S06DONE - Update published	C/R: broadcast
154:1237 / staff	S07ACTIVE - Manual fallback active	C/R: manual note extension
155:931 / staff	S08CONFLICT - Review possible duplicate	R/U: reconciliation
155:1008 / staff	S08DONE - Reconciliation complete	R/U: reconciliation
35:168 / admin	A01 - Operations overview / Mobile	R: information/navigation only
35:212 / admin	A02 - Live queue / Mobile	R: information/navigation only
35:281 / admin	A03 - Bottleneck analytics / Mobile	R: information/navigation only
35:332 / admin	A04 - Service incidents / Mobile	R/U: incident verification
35:374 / admin	A05 - Broadcast centre / Mobile	C/R/D: message publication/withdrawal extension
35:416 / admin	A06 - Reports / Mobile	R: report preview/export (export is not CRUD)
35:485 / admin	A07 - System health / Mobile	R: service health; admin status update in /manage
35:537 / admin	A08 - Audit log / Mobile	R: immutable audit
156:921 / admin	A04REVIEW - Review verification / Mobile	R/U: incident verification

Node / role	Screen	Flow operations
156:3178 / admin	A04DONE - Incident verified / Mobile	R/U: incident verification
156:3305 / admin	A05REVIEW - Review broadcast / Mobile	C/R/D: message publication/withdrawal extension
156:3432 / admin	A05DONE - Broadcast published / Mobile	C/R/D: message publication/withdrawal extension
157:1337 / admin	A06READY - Report preview / Mobile	R: report preview/export (export is not CRUD)
157:3600 / admin	A07OUTAGE - Outage procedure / Mobile	R: service health; admin status update in /manage
208:1545 / admin	ANOTIFY - Notifications / Mobile	R: information/navigation only
405:2122 / admin	AMORE - Admin tools / Mobile	R: information/navigation only
405:2123 / admin	A02PAUSED - Paused service / Mobile	R: information/navigation only
405:2124 / admin	A05LOCKED - Broadcast centre / Mobile	C/R/D: message publication/withdrawal extension
405:2125 / admin	A08PUBLISHED - Audit log / Mobile	R: immutable audit

Appendix: evidence pack and viva preparation

Evidence pack: docs/testing/functional-tests.md; usability-protocol.md; U01-U05-session.md; usability-results.json and generated usability-summary.json; traceability.json; archived docs/evidence screenshots; current build and test logs when finalized.

Each member should demonstrate their actual interface flow, identify its requirements and CRUD actions, explain the data path/authorization, show one valid and one invalid input, explain a functional defect and fix, and discuss a participant issue with a retest or planned improvement.

Patient demonstration: retrieve a linked token, interpret progress/estimate, save/revoke consent, explain missed-turn review. Staff: register/edit/select/priority/call and reconcile a reviewed note. Administrator: verify/publish/change service/report, preserving audit. Accessibility/testing member: demonstrate text/display/help and explain the real participant evidence.

Architecture trade-offs: explain why transactions do not run offline; why UI-only role checks are insufficient; why current-session averages are not historical analytics; and why an approaching turn is not a clinical call.